

PROJECT DEVELOPMENT

Visitor Management System

Introduction

The Development Phase focuses on implementing the Smart Visitor Access Management System using the ServiceNow platform. The application was developed to automate visitor registration, approval management, security validation, visitor tracking, and reporting processes.

The system was built using various ServiceNow components such as Custom Tables, Client Scripts, Business Rules, Flow Designer, UI Actions, Scheduled Jobs, Notifications, and Dashboards. The objective of the development phase was to create a secure, scalable, and automated visitor management solution that minimizes manual intervention while improving operational efficiency and security compliance.

Development Environment

Platform Used

ServiceNow Developer Instance

Development Technologies

- ServiceNow Studio
- Custom Tables
- Form Designer
- Client Scripts
- Business Rules
- Flow Designer
- Email Notifications
- UI Actions
- Scheduled Jobs
- Dashboards & Reports

Development Approach

The application was developed using a modular approach where each functional component was implemented and tested independently before integration into the complete system.

Application Development

Application Name

Smart Visitor Access Management System

Application Purpose

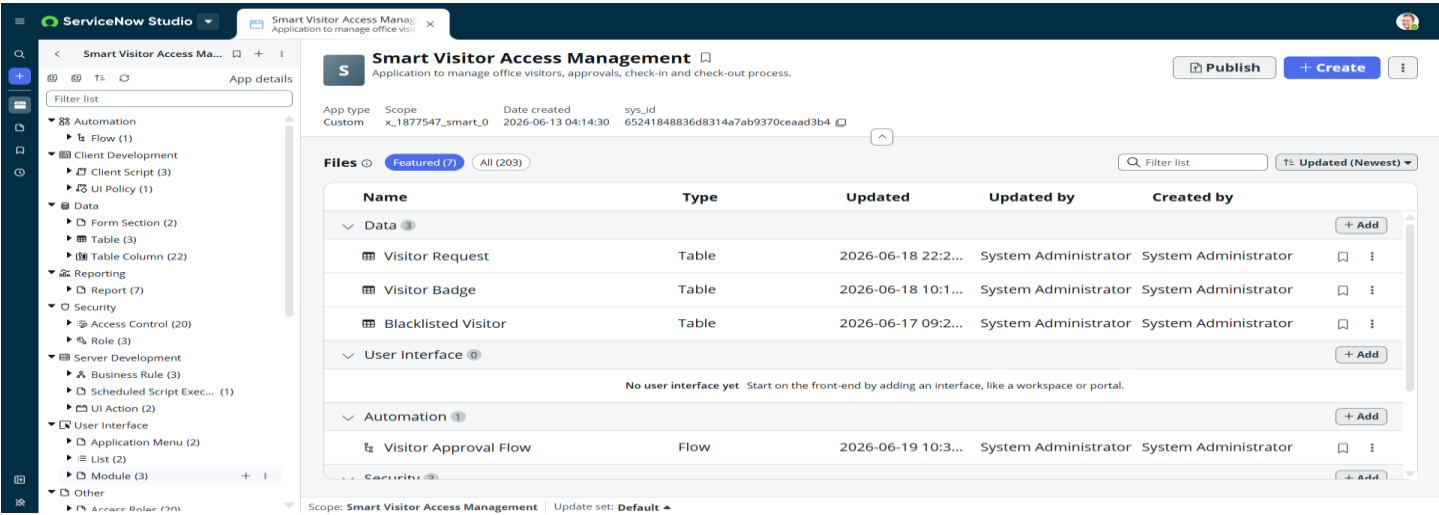
The application serves as a centralized platform for managing visitor requests from registration to exit tracking.

Core Functionalities

- Visitor Registration
- Visitor Validation
- Approval Management
- Blacklist Verification
- Security Notifications
- Visitor Check-In
- Visitor Check-Out
- VIP Visitor Management
- Dashboard Reporting

Application Benefits

- Reduced manual effort
- Improved visitor security
- Faster approval process
- Enhanced visitor monitoring
- Better audit tracking



Visitor Request Module Development

A custom table named **Visitor Request** was developed to maintain visitor information and track the complete visitor lifecycle.

Table Structure

Field Name	Description
Visitor ID	Unique Visitor Identifier
Visitor Name	Visitor Full Name
Phone Number	Visitor Contact Number
Email	Visitor Email Address
Company Name	Visitor Organization
Purpose	Reason for Visit
Visit Date	Planned Visit Date
Host Employee	Employee Being Visited
Status	Request Status
VIP Visitor	VIP Indicator
Check-In Time	Visitor Entry Time
Check-Out Time	Visitor Exit Time
Priority	Request Priority

Development Outcome

The table acts as the central repository for storing visitor information and supports all workflow operations within the application.

FavoritesHistoryWorkspacesAdmin

Table - Visitor Request

Application scope: Smart Visitor Acc...
Update set: Default [Smart Visitor A...

Table
Visitor Request

Column label

Type

Reference

Max length

Default value

Display

Sys ID

Sys ID (GUID)

(empty)

32

false

Created

Date/Time

(empty)

40

false

Priority

Choice

(empty)

40

false

Email

String

(empty)

100

false

Visitor ID

String

(empty)

40

false

Check In Time

Date/Time

(empty)

40

false

Badge Number

String

(empty)

40

false

Status

Choice

(empty)

40

false

Updated by

String

(empty)

40

false

Updates

Integer

(empty)

40

false

Company Name

String

(empty)

40

false

Host Employee

Reference

User

32

false

Purpose

Choice

(empty)

40

false

Check Out Time

Date/Time

(empty)

40

false

Number

String

(empty)

40

javascript:global.getNextObjNumberPadded();

false

Updated

Date/Time

(empty)

40

false

Visitor Name

String

(empty)

100

false

Visit Date

Date

(empty)

40

false

Created by

String

(empty)

40

false

VIP Visitor

True/False

(empty)

40

false

Form Development

A user-friendly visitor registration form was developed to simplify request submission.

Visitor Information Section

- Visitor Name
- Phone Number
- Email Address
- Company Name

Visit Information Section

- Purpose of Visit
- Visit Date
- Host Employee
- VIP Visitor Option

Features Implemented

- Mandatory Field Validation
- Structured Form Layout
- Improved User Experience
- Simplified Data Entry

Development Outcome

The form provides a standardized mechanism for capturing visitor information accurately and efficiently.

Favorites

History

Workspaces

Admin

Visitor Request - Create VIS0095

Application scope: Smart Visitor A...
Update set: Default | Smart Visitor...

Visitor Request
New record

Submit

Visitor Name

Phone Number

Email

Company Name

Purpose

-- None --

Visit Date

Badge Number

Host Employee

Status

-- None --

VIP Visitor

Check In Time

Check Out Time

Visitor ID

Number

VIS0095

Priority

-- None --

Submit

Validation Development

To ensure data integrity and security, multiple validation mechanisms were implemented.

Phone Number Validation

A Client Script was developed to validate visitor phone numbers.

Features

- Accepts only 10-digit numbers
- Rejects invalid formats
- Displays user-friendly error messages

Visit Date Validation

A Client Script was implemented to prevent invalid scheduling.

Features

- Blocks past dates
- Allows current and future dates
- Ensures valid visitor scheduling

Development Outcome

These validations significantly improve data quality and reduce incorrect visitor registrations.

The screenshot shows a web application interface for creating a visitor request. The header includes navigation links: Favorites, History, Workspaces, Admin, and a title bar 'Visitor Request - Create VIS0095'. A red error message banner at the top reads: 'Phone Number must contain exactly 10 digits'. The form fields are organized into two columns. The left column contains: Visitor Name (sas), Phone Number (13365hhngh), Email, Company Name, Purpose (-- None --), Visit Date, and Badge Number. The right column contains: Host Employee, Status (-- None --), VIP Visitor (checkbox), Check In Time, Check Out Time, Visitor ID, Number (VIS0095), and Priority (-- None --). A green 'Submit' button is located at the bottom left of the form area.

Business Rule Development

Duplicate Visitor Detection

A Business Rule was developed to prevent duplicate visitor registrations.

Functionality

- Checks existing records
- Compares phone number and visit date
- Prevents duplicate submissions

Benefits

- Eliminates redundant records
- Improves database accuracy
- Enhances visitor tracking

Blacklisted Visitor Verification

A Business Rule was implemented to identify restricted visitors.

Functionality

- Verifies visitor details against blacklist records
- Blocks unauthorized registrations
- Displays security warnings

Benefits

- Strengthens organizational security
- Prevents unauthorized access
- Supports compliance requirements

Favorites

History

Workspaces

Admin

Visitor Request - Create VIS0095

Application scope: Smart Visitor A...
Update set: Default [Smart Visitor...]

Visitor Request
New record

Submit

This visitor is blacklisted

Invalid insert

Visitor Name

sas

Phone Number

9876543210

Email

Company Name

Purpose

-- None --

Visit Date

Badge Number

Host Employee

Status

-- None --

VIP Visitor

Check In Time

Check Out Time

Visitor ID

Number

VIS0095

Priority

-- None --

Submit

Workflow Development

A workflow was developed using ServiceNow Flow Designer to automate visitor approvals.

Flow Name

Visitor Approval Flow

Workflow Trigger

Visitor Request Created

Approval Process

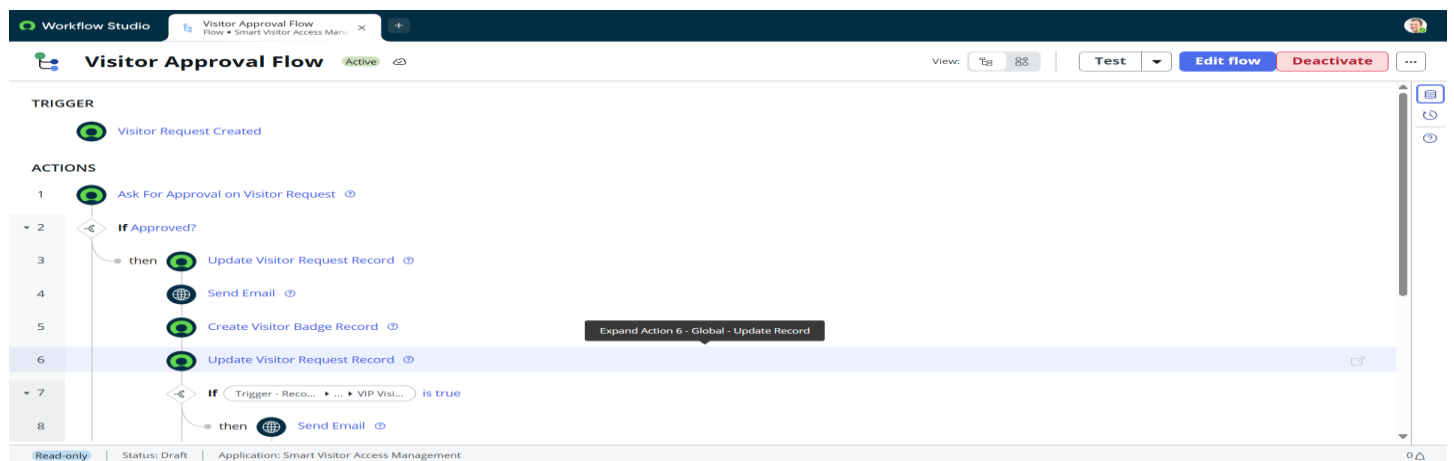
When a visitor request is submitted, the workflow automatically initiates an approval process and routes the request to the designated approver.

Workflow Activities

- Ask For Approval
- Approval Validation
- Status Update
- Security Notification
- Visitor Badge Creation
- VIP Visitor Processing

Development Outcome

The workflow eliminates manual approval handling and ensures faster processing of visitor requests.



Notification Development

Email notifications were integrated into the workflow to facilitate communication between departments.

Notification Type

Visitor Approval Notification

Notification Recipients

- Security Team
- Concerned Stakeholders

Information Included

- Visitor Name
- Visit Date
- Visit Purpose
- Host Employee

- Approval Status

Benefits

- Real-time communication
- Faster response time
- Improved operational coordination

The screenshot displays a web interface for configuring an email alert. The top navigation bar includes 'Favorites', 'History', 'Workspaces', and 'Admin'. The main header shows the email title 'Email - ALERT: VIP Visitor Arriving' with a star icon, and the application scope 'Application scope: Smart Visitor A...' with a red update message 'Update set: Default [Smart Visitor...]'.

The form is titled 'Email - ALERT: VIP Visitor Arriving - View: Sent' and includes several fields:

- Mailbox:** Outbox (with a search icon and a link icon)
- State:** Ready (with a dropdown arrow)
- Target:** (empty) (with a search icon)
- UID:** (empty text field)
- Created:** 2026-06-19 10:37:59
- Deleted:** ☐
- Weight:** 0
- Importance:** (empty text field)
- Subject:** ALERT: VIP Visitor Arriving -
- Recipients:** 717823d152@kce.ac.in
- Body:** <p><p data-tinymce-rootblock="">Add a quick alert message.</p></p> (with expand/collapse icons)
- Content type:** (empty text field)
- Headers:** Message-ID:<229839406.5.1781890679398@app129146.aus191.service-now.com>
X-ServiceNow-Source:FlowDesigner-bea482c2eb61c3548c883c2834e55c65
X-ServiceNow-Generated:true
X-ServiceNow-SysEmail-Version:2

At the top right of the form, there are 'Update' and 'Delete' buttons, along with up and down arrow icons.

Security Operations Development

Visitor Check-In Module

A custom UI Action named **Check In** was developed.

Features

- Captures visitor entry time
- Updates visitor status
- Maintains entry history

Visitor Check-Out Module

A custom UI Action named **Check Out** was developed.

Features

- Captures visitor exit time
- Updates visitor status
- Maintains exit history

Development Outcome

The module enables accurate tracking of visitor movement within the organization.

Visitor Request - VIS0080

Application scope: Smart Visitor A...
Update set: Default [Smart Visitor]

Update Check In Delete

Visitor Name: VIP Final Test Email
Phone Number: 6571023987
Email:
Company Name:
Purpose: -- None --
Visit Date:
Badge Number:

Host Employee:
Status: Approved
VIP Visitor: ☒
Check In Time:
Check Out Time:
Visitor ID:
Number: VIS0080
Priority: High

Update Check In Delete

[SN Utils] Versions (0)

VIP Visitor Management Development

A VIP visitor management feature was implemented to handle high-priority visitors.

Features

- Automatic High Priority Assignment
- VIP Identification
- Security Alert Notifications
- Priority Processing

Benefits

- Enhanced visitor experience
- Improved security readiness
- Faster approval and monitoring

Visitor Request - VIS0080

Application scope: Smart Visitor A...
Update set: Default [Smart Visitor]

Update Check In Delete

Visitor Name: VIP Final Test Email
Phone Number: 6571023987
Email:
Company Name:
Purpose: -- None --
Visit Date:
Badge Number:

Host Employee:
Status: Approved
VIP Visitor: ☒
Check In Time:
Check Out Time:
Visitor ID:
Number: VIS0080
Priority: High

Update Check In Delete

[SN Utils] Versions (0)

Automated Expiry Management

A Scheduled Job was developed to automate visitor request maintenance.

Job Name

Visitor Auto Expiry Job

Functionality

- Executes daily
- Identifies outdated approved requests
- Updates status to Expired

Benefits

- Automatic housekeeping
- Improved reporting accuracy
- Reduced administrative effort

Dashboard Development

A comprehensive dashboard was developed to monitor visitor activities and system performance.

Reports Created

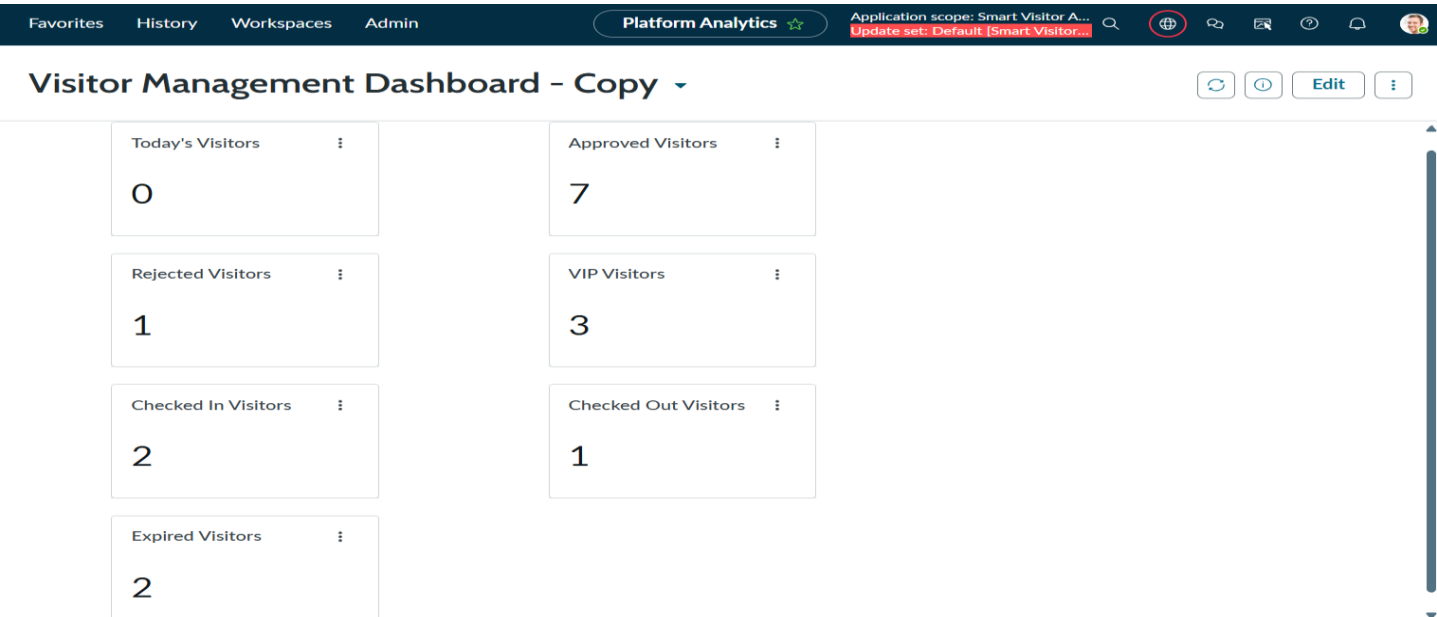
- Today's Visitors
- Approved Visitors
- Rejected Visitors
- Checked-In Visitors
- Checked-Out Visitors
- VIP Visitors
- Expired Visitors

Dashboard Features

- Real-time analytics
- Visual reporting
- Visitor trend monitoring
- Operational insights

Development Outcome

The dashboard provides management and security teams with complete visibility into visitor operations.



Testing and Validation

All developed modules were tested to verify functionality and performance.

Testing Activities

- Visitor Registration Testing
- Validation Testing
- Approval Workflow Testing
- Notification Testing
- Check-In/Check-Out Testing
- VIP Visitor Testing
- Dashboard Testing

Result

All modules performed successfully and met the intended business requirements.

Conclusion

The Development Phase successfully implemented the Smart Visitor Access Management System using the ServiceNow platform. The application integrates visitor registration, validation mechanisms, approval workflows, blacklist verification, security operations, VIP visitor handling, automated notifications, and dashboard analytics into a unified solution.

The developed system significantly improves visitor management efficiency, enhances organizational security, reduces manual effort, and provides real-time visibility into visitor activities. The solution demonstrates the effective use of ServiceNow technologies in building a scalable and enterprise-grade visitor management platform.